4190.428 Introduction to Machine Learning

Linear classifiers

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Overview

- Classification
- Learning algorithm
  - Evaluating a learning algorithm
- Linear classifiers
  - Learning linear classifiers

Classification

- A binary **classifier** is a mapping from $\mathbb{R}^d \to \{-1, +1\}$.
- Use the letter h (for hypothesis) to stand for a classifier
- Classification process : $\mathbf{x} \to h \to y$

- Define function $\varphi(\mathbf{x})$, whose domain is $\mathbb{R}^d$, where φ represent **features** of $\mathbf{x}$
- Let $h : \varphi(\mathbf{x}) \to \{-1, +1\}$

Classification

- In **supervised learning,** training data set of the form is

$$\mathscr{D}_n = \{(\mathbf{x}^{(1)}, y^{(1)}), \ldots, (\mathbf{x}^{(n)}, y^{(n)})\}$$

- Assume that each $\mathbf{x}^{(i)}$ is a $d \times 1$ **column vector**.
- The intended meaning of data :
  - Given an input $\mathbf{x}^{(i)}$, the learned hypothesis should generate output $y^{(i)}$.

Classification

- For making classifier useful, it works well on **new** data.
- So we have to assume a connection between the training data and testing data.
- typically they are drawn independently from the same probability distribution.

- Define the **training error of h** to be

$$\epsilon_n(h) = \frac{1}{n}\sum_{i=1}^{n}\begin{cases}1, & \text{if } h(\mathbf{x}^{(i)}) \neq y^{(i)} \\ 0, & \text{otherwise}\end{cases}$$

- For **generalizing well** to new data, has a small **test error**

$$\epsilon(h) = \frac{1}{n'}\sum_{i=n+1}^{n+n'}\begin{cases}1, & \text{if } h(\mathbf{x}^{(i)}) \neq y^{(i)} \\ 0, & \text{otherwise}\end{cases}$$

Learning algorithm

- A hypothesis class $\mathscr{H}$ is a set of possible classifiers
- A binary **classifier** is a mapping from $\mathbb{R}^d \to \{-1, +1\}$.

- Learning algorithm :

$$\mathscr{D}_n \longrightarrow \text{learning alg } (\mathscr{H}) \longrightarrow h$$

- Choice of $\mathscr{H}$ can have a big impact on the test error of the h
- One way to get h that generalizes well is to restrict the size, or **expressiveness** of $\mathscr{H}$

Linear classifiers

- A linear classifier is defined by a vector of parameters $\boldsymbol{\theta} \in \mathbb{R}^d$, scalar $\theta_0 \in \mathbb{R}$.
- So, the hypothesis class $\mathscr{H}$ is the set of all vectors in $\mathbb{R}^{d+1}$.

- The Classifier:

$$h(\boldsymbol{x}; \boldsymbol{\theta}, \boldsymbol{\theta_0}) = \text{sign}(\boldsymbol{\theta}^\top \boldsymbol{x} + \boldsymbol{\theta_0}) = \begin{cases} +1 & \text{if } \boldsymbol{\theta}^\top \boldsymbol{x} + \boldsymbol{\theta_0} > 0 \\ -1 & \text{otherwise} \end{cases}$$

  - $\boldsymbol{x}$ and $\boldsymbol{\theta}$ are both d $\times$ 1 column vectors.
  - So $\boldsymbol{\theta}^\top \boldsymbol{x}$ is 1 $\times$1, which in math is the same as a scalar

Linear classifiers

- Think of $\boldsymbol{\theta}, \boldsymbol{\theta}_{\mathbf{0}}$ as specifying a hyperplane.
- It divides $\mathbb{R}^d$, the space our $\mathbf{x}^{(i)}$ points live in, into half-spaces.

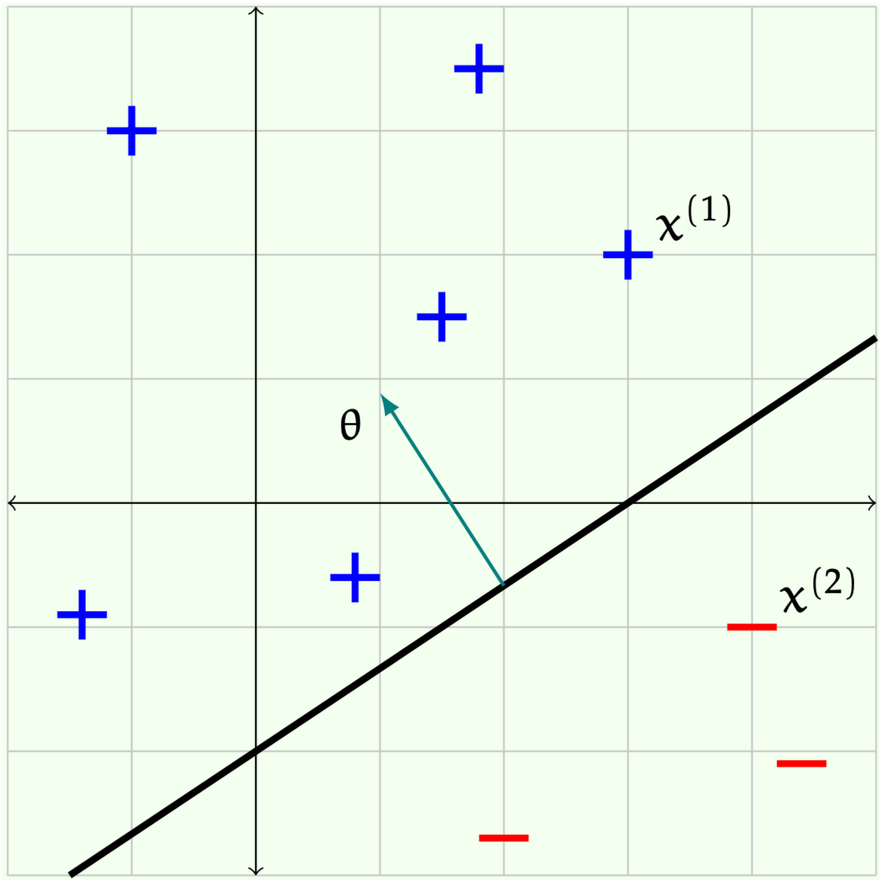
- The one that is on the same side as the normal vector is the **positive** half-space and all points in that space as positive.
- The half-space on the other side is **negative** and all points in it are classified as negative.

Linear classifiers

- Example:
  - Let h be the linear classifier defined by $\boldsymbol{\theta} = \begin{bmatrix} -1 \\ 1.5 \end{bmatrix}$, $\theta_0 = 3$
  - The diagram below shows several points classified by h
  - In particular, let $\boldsymbol{x}^{(1)} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ and $\boldsymbol{x}^{(2)} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$.

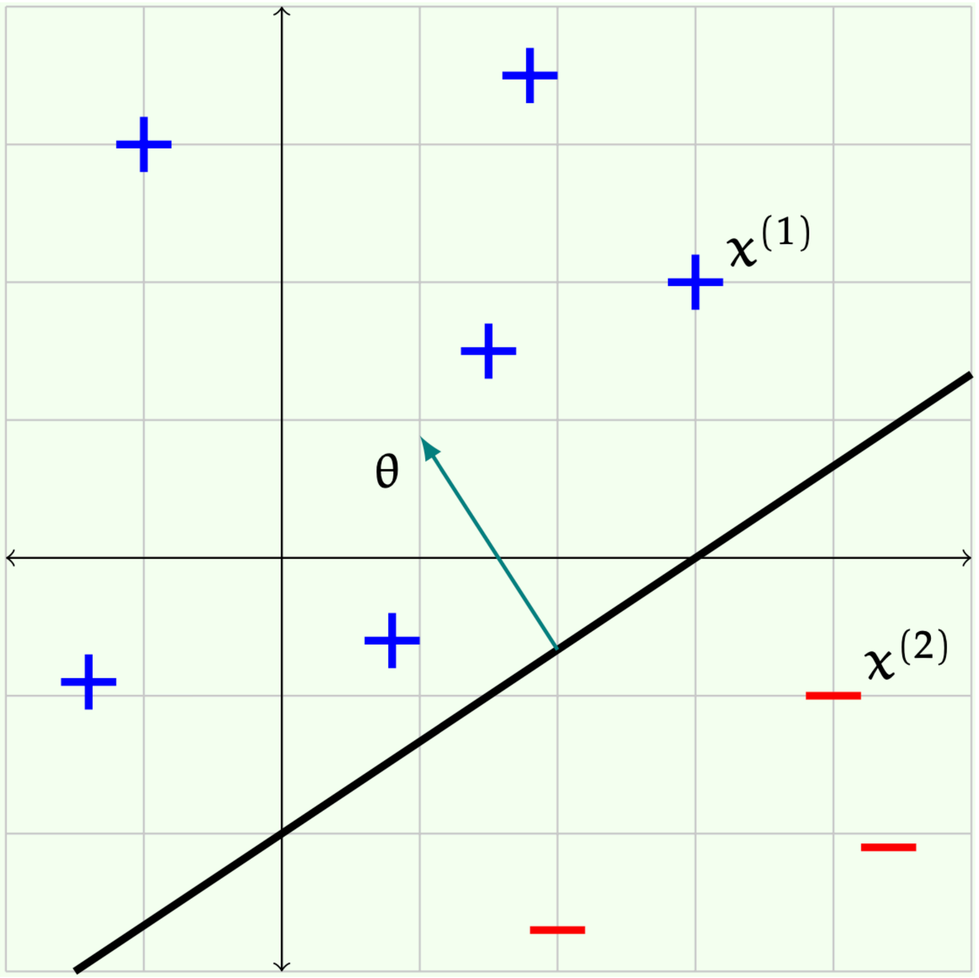
$$h(\boldsymbol{x}^{(1)}; \boldsymbol{\theta}, \theta_0) = \text{sign}\left(\begin{bmatrix} -1 & 1.5 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} + 3 \right) = \text{sign}(3) = +1$$

$$h(\boldsymbol{x}^{(2)}; \boldsymbol{\theta}, \theta_0) = \text{sign}\left(\begin{bmatrix} -1 & 1.5 \end{bmatrix} \begin{bmatrix} 4 \\ -1 \end{bmatrix} + 3 \right) = \text{sign}(-2.5) = -1$$



Linear classifiers

- **Question**:
  - What is the green vector normal to the hyper plane? Specify it as a column vector.
  - What change would you have to make to $\boldsymbol{\theta}, \boldsymbol{\theta_0}$ if you wanted to have the separating hyperplane in the same place, but to classify all the points labeled '+' in the diagram as negative and all the points labeled '-' in the diagram as positive?



Learning linear classifiers

- Consider a very simple learning algorithm.
  - The idea is to generate k possible hypotheses by generating their parameter vectors at random.

Algorithm 1: Random-Linear-Classifier($\mathcal{D}_n$, k, d)

for $j = 1$ *to* k **do**
  Randomly sample $\left(\boldsymbol{\theta}^{(j)}, \theta_0^{(j)}\right)$ from $(\mathbb{R}^d, \mathbb{R})$
end
$j^* = \arg\min_{j \in \{1, \cdots, k\}} \epsilon_n \left(\boldsymbol{\theta}^{(j)}, \theta_0^{(j)}\right)$
return $\left(\boldsymbol{\theta}^{(j^*)}, \theta_0^{(j^*)}\right)$

Learning linear classifiers

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- Study Question:
  - What do you think happens to $\mathscr{E}_n(h)$, where h is the hypothesis returned by RANDOM-LINEAR-CLASSIFIER, as k is increased?
  - What properties of $\mathscr{D}_n$ do you think will have an effect on $\mathscr{E}_n(h)$?

Evaluating a learning algorithm

- There are many potential sources of variability in the possible result of computing test error on a learned hypothesis h.
  - Which particular **training examples** occurred in $\mathscr{D}_n$
  - Which particular **testing examples** occurred in $\mathscr{D}_{n'}$
  - Randomization inside the learning **algorithm** itself

- We would like to execute the following process multiple times:
  - Train on a new training set
  - Evaluate resulting h on a testing set **that does not overlap the training set**

Evaluating a learning algorithm

- We can re-use data with **cross validation.** But it's harder to do theoretical analysis.

Algorithm 2: Cross-Validate($\mathcal{D}$, k)

```
divide D into k chunks D_1, D_2, ..., D_k (of roughly equal size)
for i = 1 to k do
    train h_i on D \ D_i (withholding chunk D_i)
    compute "test" error ε_i(h_i) on withheld data D_i
end
return (1/k) Σ_{i=1}^{k} ε_i(h_i)
```

- It's very important to understand that cross-validation neither delivers nor evaluates a single particular hypotheses h.
- It evaluates the **algorithm** that produces hypotheses.

QnA

- Any questions?
- Note: this lecture notes are largely adopted from
  - MIT 6.036(2019 Fall), Stanford CS229(2023-2024 Summer)